

Model Questions and Answers for LASER and Fiber Optics

1. What is LASER? Discuss its different characteristics.

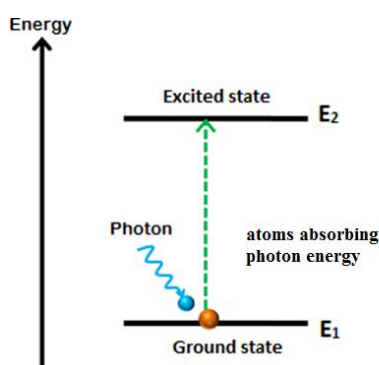
Ans. A laser is a device that produces a light beam with some remarkable properties, viz.

1. The light is nearly monochromatic.
2. The light is coherent (temporal as well as spatial), with the waves all are exactly in phase with each other.
3. A laser beam hardly diverges.
4. The beam is extremely intense (large energy density), more than the light from any other source.
5. Highly collimated beam.

2. Explain absorption, spontaneous emission and stimulated emission in the context of laser with suitable diagrams. Which one is the most important step for laser production?

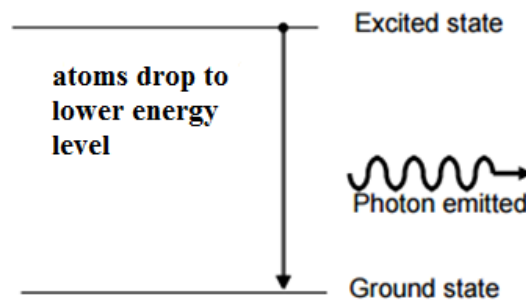
Ans. Absorption

Every atom, according to the quantum theory, can reside only in certain discrete energy states or energy levels. Normally, the atoms are in the lowest energy state or ground state. When light with a photon of energy $h\nu = E_2 - E_1$ falls on a substance, the atoms in the ground state (E_1 , say) can be excited to go to one of the higher levels (E_2 , say). This process is called absorption.



Spontaneous emission

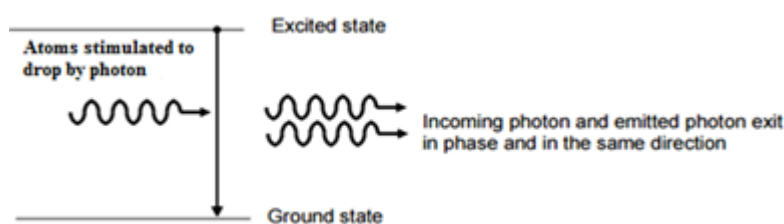
Consider an atom (or molecule) of the material staying initially in an excited state E_2 . Since $E_2 > E_1$, the atom will tend to spontaneously decay to the ground state E_1 to attain the lowest energy state, and a photon of energy $h\nu = E_2 - E_1$ is released in a random direction as shown below. No external radiation is required to initiate the emission. This process is called “spontaneous emission”.



When the released energy [equal to the energy difference $E_2 - E_1$] is delivered in the form of an electromagnetic wave, the process called "radiative emission" which is one of the two possible ways. "Non-radiative" decay is occurred when the energy difference ($E_2 - E_1$) is delivered in some form other than electromagnetic radiation (e.g. it may be transferred to the kinetic energy of the surrounding).

Stimulated emission

Stimulated Emission requires the presence of external radiation. When an incident photon of energy $h\nu = E_2 - E_1$ passes by an atom in an excited state E_2 , it stimulates the atom to drop or decay to the lower state E_1 . This is due to the fact that EM field, with proper frequency, will exert an oscillating force on the excited atom. In this process, the atom releases a photon of the same energy, direction, phase and polarization as that of the photon passing by, the net effect is two identical photons (i.e. energy = $2h\nu$) in the place of one, or an increase in the intensity of the incident beam. It is precisely this process of stimulated emission that makes possible the amplification of light in lasers.



Because of the fact that photon multiplication or amplification takes place in case of stimulated emission, it is the most important and key step for laser production.

3. Write short note on stimulated emission explaining its importance for laser production.

Ans. Answer same as the last part (i.e. stimulated emission) of Q. 2.

4. State Boltzmann's distribution law and hence show that in normal conditions ground state will remain most populated.

Ans. Boltzmann's law states that if an assemblage of atoms is in state of thermal equilibrium at an absolute temp. T , the number of atoms N_2 in one energy level E_2 is related to the number N_1 in another energy level E_1 by the equation

$$N_2 = N_1 e^{-(E_2 - E_1)/KT}$$

In the above equation if $E_2 > E_1$, thus clearly $N_2 < N_1$. (K is Boltzmann's constant = 1.38×10^{-23} J/K and T is the absolute temperature in Kelvin). Or, in other words, the number of atoms decreases as the energy level increases. As the temp increases, more atoms will attain higher energy levels. However, the ground energy level or the ground state will remain most populated.

5. Explain Einstein's A and B coefficients in relation with the theory of lasing.

Ans. Let us consider a two level atomic system in state of thermal equilibrium at an absolute temp. T . Let the number of atoms N_2 are in one energy level E_2 and the number N_1 are in another energy level E_1 ($E_2 > E_1$). An atom in level E_2 can decay to level E_1 by emission of photon. Let us call A_{21} the transition probability per unit time for spontaneous emission from level E_2 to level E_1 . Then the number of spontaneous decays per second is $N_2 A_{21}$, i.e. the number of spontaneous decays per second = $N_2 A_{21}$ (as it will depend only on population N_1)

In addition to these spontaneous transitions, there will be induced or stimulated transitions. The total rate of these induced transitions between level 2 and level 1 is proportional to the population N_2 and the density (U_ν) of radiation of frequency ν , where

$$\nu = (E_2 - E_1)/h, \quad h \text{ being the Planck's constant.}$$

Let B_{21} and B_{12} denote the proportionality constants for stimulated emission and absorption. Then number of stimulated downward transition in stimulated emission per second = $N_2 B_{21} U_\nu$.

Similarly, the number of stimulated upward transitions per second = $N_1 B_{12} U_\nu$, as it will depend on the population N_1 of the lower level and the density (U_ν) of radiation of frequency ν . The proportionality constants A and B are known as the Einstein A and B coefficients.

6. What is meant by metastable state? What is its significance?

Ans. The atoms remain in the excited state for a very short period of time $\sim 10^{-8}$ sec, after that they spontaneously release the excess energy. For stimulated emission to occur, the atoms should stay longer (typically $\sim 10^{-4}$ sec) in the excited state to compensate for the loss in population due to spontaneous emission till the condition $N_2 > N_1$ gets satisfied (N_2 being the

population at higher energy level). Such relatively long-lived states with life time $\sim 10^{-3}$ sec are called metastable (temporarily stable) states and their energy lie between upper excited state and the lower state.

The significance of metastable state is that the population of atoms in such a state is greater than that in lower state, which leads to population inversion.

7. Explain population inversion, active system and pumping.

Ans. The term population inversion describes an assembly of atoms in which the majority of atoms are in energy levels above the ground state; normally the ground state is occupied to the greatest extent as per the Boltzmann distribution law $N_2 = N_1 e^{-(E_2 - E_1)/KT}$

Significance of population inversion is that if it exists, then the number of stimulated emission will be higher and a light beam will increase in intensity i.e. it will be amplified as it passes through the medium.

A system in which population inversion has been achieved is called an **active system**. An active system is one that can produce laser beam.

To attain and sustain population inversion, the atoms in the material must be continuously excited from the lower energy level to the higher energy level. This is done by continuous supply of energy to the medium. The process is known as pumping. Popular pumping schemes include optical and electrical pumping.

8. What is meant by pumping? Why it is necessary for laser production? Discuss different types of pumping mechanisms.

Ans. To attain and sustain population inversion, the atoms in the material must be continuously excited from lower energy level to higher energy level. This is done by continuous supply of energy to the medium. The process is known as pumping.

Pumping is necessary for laser production as it helps to achieve and maintain population inversion, the key condition for producing laser.

Different types of pumping schemes are as follows.

i) OPTICAL PUMPING

Here optical energy in the form of photon with right frequency is used to excite the atoms in the medium. For supplying optical energy, a discharge tube is employed to excite atoms from the ground state to a higher or excited state. Optical pumping is suitable for any medium which is transparent to light.

ii) ELECTRICAL PUMPING

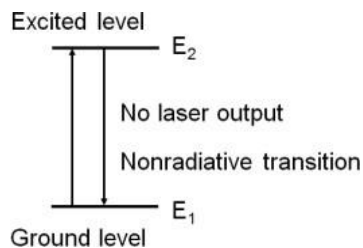
It is used for some mediums which can conduct electricity without affecting laser activity. Suitable for gas and semiconductor lasers. In gaseous medium, the gas is ionized by a pulse of high electric voltage, current flowing through the gas provides necessary energy to excite the atoms to uppermost level from where they decay to metastable level.

9. Explain why simple heating cannot achieve population inversion.

Ans. By heating the material the state of population inversion cannot be achieved. By heating the material, the average energy of the atoms is increased (i.e. overall system is shifted to the higher energy level) without achieving the required condition for population inversion, i.e. $N_2 > N_1$ (considering N_2 is the population at higher energy level). Therefore, to achieve the condition different methods are used.

10. Why lasing cannot be obtained using only two energy levels?

Ans. It is not possible to achieve population inversion with a 2-state (level) system. If the incoming radiation flux is made very large the probability of stimulated emission and absorption can be made to far exceed the rate of spontaneous emission. But in 2-state system, the best we can get is $N_1 = N_2$.



Suppose there are only 2 levels, a metastable state, corresponding to energy $h\nu$, above the ground state. The more photons with frequency ν we pump into the assembly of atoms, the more upward transitions there will be from the ground state to the metastable state. However, pumping will also stimulate downward transitions from metastable state to the ground state. i.e. population and depopulation processes take place simultaneously.

In other words, the incoming flood of photons is just as likely to cause an atom to drop (stimulated emission) as it is to cause an atom to rise (absorption). When half the atoms are in each state, the rate of stimulated emissions will be equal the rate of stimulated absorption, so the assembly can't have more than half its atoms in metastable state. In this situation laser amplification cannot occur.

11. Distinguish between 3-level and 4-level lasers.

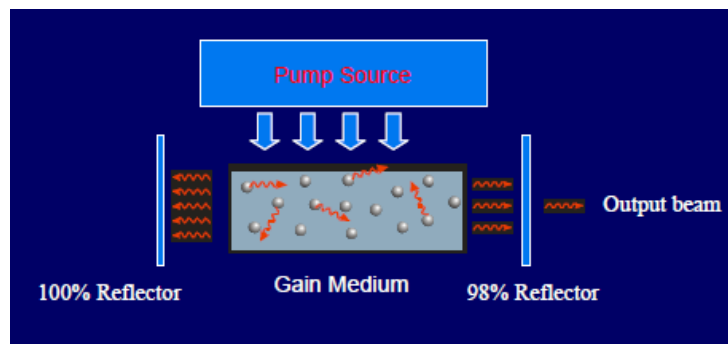
Ans. Compared to 3-level lasers 4-level lasers have several advantages like

- The lasing threshold of a four level laser is lower.

- The efficiency is higher.
- Required pumping rate is lower.
- Continuous operation is possible in four level laser (e.g. He-Ne laser).
Threelevel laser gives pulsed operation (e.g. Ruby laser)

12. What is an optical resonator? Explain in detail how it plays a key role in laser production.

Ans. In laser, active medium is an amplifier, which is converted into an oscillator through the feedback mechanism established by an optical resonator. A pair of optically plane parallel mirrors, enclosing laser medium in between them, is known as an **optical resonant cavity**. One of the mirror is partially reflecting and the other is fully reflecting. The laser oscillation is initiated by photons spontaneously emitted by some of the excited atoms. The photons emitted along the optic axis of the resonant cavity travel through the medium and trigger stimulated emissions. They are reflected by the end mirror and reverse their path. The optical resonator selects the direction in which the light is to be amplified; the direction being the optical axis of the pair of mirrors. Thus, the optical cavity makes the laser beam directional. The photons, thus fed back into the medium, travel toward the opposite end mirror causing more stimulated emissions. It is again reflected back from that end. As the process repeats itself substantial light amplification takes place.



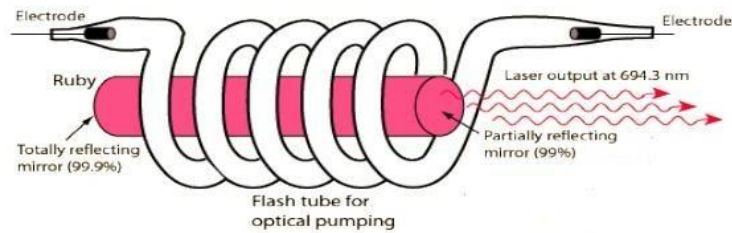
In order to make the stimulated emission dominate spontaneous emission, a high radiation density is required to be present in the active medium. The optical cavity builds up the photon density to a very high value through repeated reflection of photons and confines them within the medium.

Optical cavity selects and amplifies only certain frequencies causing the laser output to be highly monochromatic.

13. Explain the construction and working of Ruby lasers with energy level diagrams.

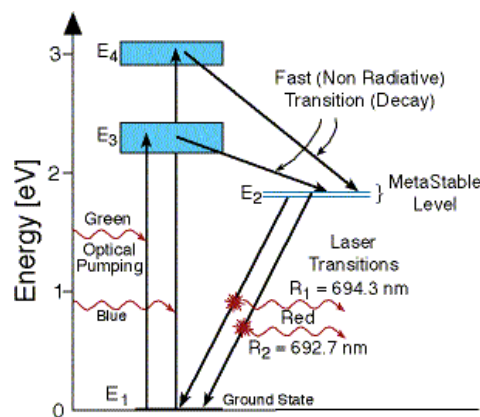
Ans. Ruby belongs to the family of gems consisting of Al_2O_3 with various types of impurities. For example, pink Ruby contains 0.05% Cr atoms. The construction of ruby laser is given

below.



The ruby laser consists of a ruby rod made of chromium doped ruby material. At the opposite ends of this rod there are two silver polished mirrors. Whose one is fully polished and other is partially (~ 98%) polished. A spring is attached to the rod with fully polished end for adjustment of wave length of the laser light. Around the ruby rod a flash light is kept for the pump input. The whole assembly is kept in the glass tube. Around the neck of the glass tube the R.F source and switching control is designed in order to switch on and off the flash light for desired intervals.

When the R.F is switched on a flash of light is obtained around the ruby rod. This causes the Cr^{3+} ions within ruby rod to move from lower energy band E_1 towards higher E_3 and E_4 bands by the green and blue components of the white light. The Cr^{3+} ions undergo non-radiative transitions from these energy levels to level E_2 , which is a metastable state. Population inversion takes between E_2 and the ground state E_1 , making these two levels suitable for lasing action. Transitions between the levels E_2 and E_1 result in emission of the laser light of wavelength 693.4 nm. The energy level diagram is given below.



Advantages of Ruby Laser

1. The ruby lasers are economical.
2. Since the ruby is in solid form therefore there is no chance of wasting material of activemedium.

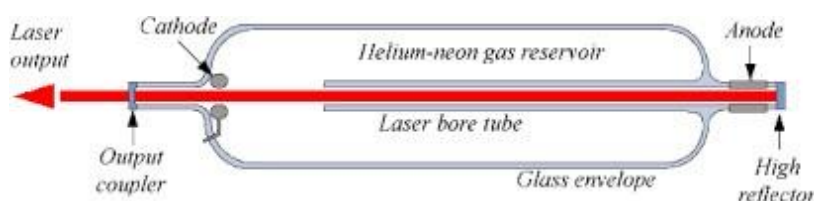
Disadvantages of Ruby Laser

1. In ruby lasers no significant stimulated emission occurs, until at least half of the groundstate electrons have been excited to the Meta stable state.

2. Efficiency of ruby laser is comparatively low.
3. Optical cavity of ruby laser is short as compared to other lasers, which may be considered a disadvantage.

14. Explain the construction and working of He-Ne lasers with energy level diagrams.

Ans. A helium-neon laser, usually called a He-Ne laser, is a type of small gas laser. It is a four-level laser. Its usual operation wavelength is 632.8 nm, in the red portion of the visible spectrum. It operates in Continuous Working (CW) mode.



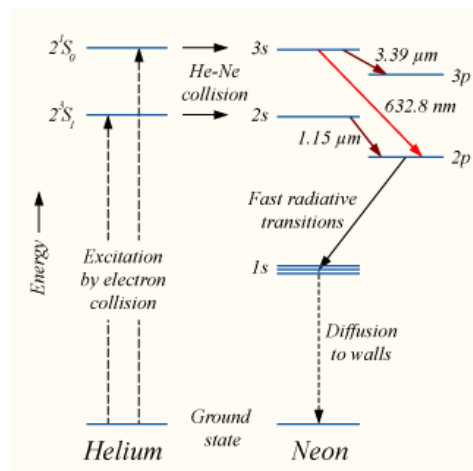
The basic construction of a He-Ne laser is shown above. The setup consists of a discharge tube of length 80 cm and bore diameter of 1.5 cm. A mixture of helium and neon gases, in approximately a 10:1 ratio, contained at low pressure in a glass envelope. The energy or pumping of the laser is provided by a high voltage electrical discharge passed through the gas between the anode and cathode within the tube. The optical cavity of the laser usually consists of two concave mirrors or one plane and one concave mirror, one having very high (typically 99.9%) reflectance and the output coupler mirror allowing approximately 1% transmission.

In order to pump the ground state Ne atoms to the excited state, the ground state atoms of He are pumped to the 2^3S_1 and 2^1S_0 long-lived, metastable excited states by passing an electric discharge. The excited He atoms then collide with the ground state Ne atoms and transfer their energy to the Ne atoms exciting them into the 2s and 3s states. This is due to a coincidence of energy levels between the helium and neon atoms.

This process is given by the reaction equation:



where (*) represents an excited state, and ΔE is the small energy difference between the energy states of the two atoms, of the order of 0.05 eV. The typical energy diagram is shown below.



The reason behind the above indirect transfer of energy to Ne is the small atomic cross-section of Ne which makes direct energy transfer a bit difficult to realize.

The number of neon atoms entering the excited states builds up as further collisions between helium and neon atoms occur, causing a population inversion between the neon $3s$ and $2s$, and $3p$ and $2p$ states. Spontaneous emission between the $3s$ and $2p$ states results in emission of

632.8 nm wavelength light, the typical operating wavelength of a He-Ne laser.

After this, fast radiative decay occurs from the $2p$ to the $1s$ energy levels, which then decay to the ground state via collisions of the neon atoms with the container walls. Because of this last required step, the bore size of the laser cannot be made very large and the He-Ne laser is limited in size and power.

Typical power output for He-Ne lasers lie between 1 and 50 mW for continuous wave operations for inputs of $\sim 5\text{-}10 \text{ W}$.

Advantages of He-Ne lasers

1. He-Ne laser is a continuous wave laser.
2. Compared to solid state lasers He-Ne laser is more directional and much more monochromatic.

Disadvantage of He-Ne lasers

1. Since this type of lasers use internal mirrors (i.e. with mirrors sealed inside the discharge tube), mirrors are very easily eroded by gas discharge and have to be replaced.

15. List the applications of laser in different fields.

Ans. Following are some of the applications of the LASER.

1. The narrow red beam of He-Ne laser is used in supermarkets to read bar codes.
2. Measuring distances
3. Red He-Ne laser have many industrial and scientific uses. They are widely used in laboratory demonstrations.
4. Laser is used in guided smart weapon.
5. Because of the high power levels available (combined with reasonable cost for the laser), CO₂ lasers are frequently used in industrial applications for cutting and welding, while lower power level lasers are used for engraving. They are also very useful in surgical procedures because water (which makes up most biological tissue) absorbs this frequency of light very well. Some examples of medical uses are laser surgery, skin resurfacing etc.
6. Semiconductor lasers are used in DVD and CD players, as Barcode readers, in telecommunications (via optical fibres), Image scanning etc.
7. Holography.